

Sunyoung Jung

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Education

Mar. 2025 ~	Yonsei University Department of Artificial Intelligence <i>Advisor: SeongJae Hwang</i> Ph.D.	Seoul, Republic of Korea
Mar. 2023 ~ Feb. 2025	Yonsei University Department of Electrical and Electronic Engineering <i>Advisor: Dong-Hyun Kim</i> M.S	Seoul, Republic of Korea
Mar. 2021 ~ Feb. 2023	Yonsei University Department of Biotechnology <i>B.S. in Biotechnology</i>	Seoul, Republic of Korea
Mar. 2019 ~ Feb. 2023	Yonsei University Division of Medical Engineering <i>Advisor: Sejung Yang</i> <i>B.S. in Bio Medical Engineering</i>	Wonju, Republic of Korea

Research Interests

- Video Generation
- Diffusion model
- Generative model
- Medical image processing

Conferences

1. **Sunyoung Jung***, Jiwoo Park*, Yeonkyung Lee, Tae Eun Choi, and Seong Jae Hwang, "AQUA: Aligned Query Fusion for Reference-Unbiased and Temporally Consistent Video Motion

Transfer”, Under Review

2. Jinyeong Kim*, Junhyeok Kim*, Yumin Shim, Joohyeok Kim, **Sunyoung Jung**, Seong Jae Hwang, “Interpreting vision transformers via residual replacement model”, The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), San Diego, USA – Poster
3. **Sunyoung Jung**, Yoonseok Choi, Gayoon Choi, Dong-Hyun Kim, and Seong Jae Hwang, “Refined Residual Diffusion Model for Motion Correction and Enhanced Anatomical Structures”, Preprint
4. Yoonseok Choi, **Sunyoung Jung**, Mohammed A. Al-masni, Ming-Hsuan Yang and Dong-Hyun Kim, “TESLA: Test-time Reference-free Through-plane Super-resolution for Multi-contrast Brain MRI”, MICCAI, Daejeon, Republic of Korea (Sep. 2025) – *Oral*
5. Changmin Ryu, **Sunyoung Jung**, Na-Young Shin, and Dong-Hyun Kim. “Attention-guided deep learning model focusing on myelination for predicting brain age using multi contrast MRI.” International Society for Magnetic Resonance in Medicine, Hawaii (May. 2025) – *Oral*
6. Changmin Ryu, **Sunyoung Jung**, Yoonseok Choi, and Dong-Hyun Kim. “Improving subcortical segmentation in brain MRI using knowledge distillation to enhance robustness against motion artifacts.” International Society for Magnetic Resonance in Medicine, Hawaii (May. 2025) – *Poster*
7. Yoonseok Choi, **Sunyoung Jung**, Mohammed A. Al-masni, Daniel Kim, and Dong-Hyun Kim, “SUPREM: A Super-Resolution Network for Through-Plane Structure Enhancement Using Disentangled Representation Learning in Multi-Contrast MRI”, International Society for Magnetic Resonance in Medicine, Hawaii (May. 2025) – *Poster*
8. Changmin Ryu, **Sunyoung Jung**, Yoonseok Choi, and Dong-Hyun Kim, “Improving subcortical segmentation in brain MRI using knowledge distillation to enhance robustness against motion artifacts.”, International Congress on MRI (ICMRI), Seoul, Republic of Korea (Nov. 2024) (*Best Trainee Scientific Awards Poster Presentation(Gold)*) – *Poster*
9. **Sunyoung Jung**, Yoonseok Choi, Mohammed Al-masni, Min-Young Jung, Dong-Hyun Kim, "Deformation-Aware Segmentation Network Robust to Motion Artifacts for Brain Tissue Segmentation using Disentanglement Learning.", MICCAI, Marrakesh, Morocco (Oct. 2024) – *Poster*
10. **Sunyoung Jung**, Yoonseok Choi, Mohammed A Al-masni, Dong-Hyun Kim, "Brain Tissue Segmentation robust to motion artifacts using Deformation-Aware Network", International Society for Magnetic Resonance in Medicine, Singapore (May. 2024) – *Oral*
11. Kyu-Jin Jung, Chuanjiang Cui, Jae - Hun Lee, Jun-Hyeong Kim, Kyoung-Jin Park, SooHyoung Lee, **Sunyoung Jung**, Dong-Hyun Kim, "Investigation of electrical conductivity changes during functional activity of the brain via phase - based MR - EPT: Preliminary observation", International Society for Magnetic Resonance in Medicine, Canada (Jun. 2023) – *Oral*
12. Daniel Kim, Jae-Hun Lee, Mohammed A. Al-masni, Jun-ho Kim, Yoonseok Choi, Eun-Gyu Ha, **Sunyoung Jung**, Dong-Hyun Kim, "3D Lesion Generation Model considering Anatomic Localization to Improve Object Detection in Limited Lacune Data", International Society for Magnetic Resonance in Medicine, Canada (Jun. 2023) – *Poster*

Publications

1. **Sunyoung Jung**, Yoonseok Choi, Mohammed Al-masni, Min-Young Jung, Dong-Hyun Kim, "Motion Artifact-Informed Brain Tissue Segmentation Network via Disentanglement Learning.", Under Review

2. Seuk Kim*, Mohammed A. Al-masni*, Seul Lee, **Sunyoung Jung**, Kyu-Jin Jung, Chuanjiang Cui, Sung-Min Gho, Young Hun Choi, and Dong-Hyun Kim, " Unsupervised learning for motion correction and assessment in brain magnetic resonance imaging using severity-based regularized cycle consistency.", Engineering Applications of Artificial Intelligence, Volume 142, 15 February 2025, 109978 (IF: 8.0)

Awards and Honors

- ISMRM 2024 stipend, Korean Society of Magnetic Resonance in Medicine, Korea (Mar. 2024)
- Academic Achievement Award, Yonsei University, Korea (Aug. 2019)

Research Experiences

- Undergraduate Research Student at Human Brain Function Laboratory, Seoul National University College of Medicine, Korea (Jun. 2021 ~ Sep. 2021)
- Undergraduate Research Student at Medical Imaging Lab, Yonsei University, Electrical & electronic engineering, Korea (Aug. 2022 ~ Feb. 2023)

Other Experiences

- Intern Student at Medical Equipment Management, Yonsei University Yongin Severance Hospital, Korea (Jul. 2022 ~ Aug. 2022)